

P100-066: Replication Data for : Boosting hydrogen storage and release in MOF-5 / graphite hybrids via in situ synthesis

Principia-100 | Standardized portfolio | 1 October 2026

Scenario and evaluation contract

Domain: Hydrogen storage. The current default task predicts Excess hydrogen uptake in wt.%. Independent unit: whole material composition across temperatures and both branches. Its packaged cohort contains 58 assigned rows in 1 groups.

All supplied outcomes are now exposed. Historical reserved evaluations remain part of the evidence record; future scores are retrospective. No newly established fundamental law or measured deployment impact is admitted by this packaging.

Current portfolio findings

Reproduction: The original displaced-gas term improves a weak monotone adsorption baseline but fails an optimum-pressure interpretation.

Limit: Idealized pressure subtraction is insufficient at cryogenic high pressure; fitted uptake and derived total-uptake/reuse quantities are not independent confirmation.

Retrospective extension: Correct real-gas excess modeling gives much better local curves, but improvement over a repaired strong ideal-excess baseline is only 3.19% on EG 5.

Limit: Effective pore volume is fitted from uptake, not independently measured .77 K_{peak} differs 3.91 bar from sampled maximum, but 160/273 K_{maxima} lie at upper pressure boundary and are not interior optima. PSDREADME/filename material mismatch prevents independent pore calibration.

Versioned tasks

Task family	Target and information	Source scope
original	Excess hydrogen uptake (wt.%)	original corpus
continuation	Excess hydrogen uptake (wt.%)	original corpus

Different targets, calibration budgets and cohorts have different task IDs. Their raw error values cannot be pooled into a scenario ranking. Exact equations, coefficients, permitted variables, timing, groups and per-model results are bound in the task packages.

Evaluation: the shared CLI provides numerical scoring, explicit coverage and abstention, uncertainty and event diagnostics. Agent review separately assesses scientific support; no aggregate discovery score is produced.

Detailed reference note: Hydrogen adsorption: corrected excess response and bounded peaks

Research edition: 1 October 2026. Historical results remain preserved; this reader edition uses the shared benchmark evaluator.

Supported findings and corrected physical model

The native target is excess hydrogen uptake in wt.%, not absolute storage capacity. Three compositions EG0/EG1/EG10 supply development; whole EG5 is an exposed diagnostic. Entire temperature and adsorption/desorption curves remain together. Independent normal hydrogen density and fugacity come from the Leachman equation of state through CoolProp 7.2.0; the frozen lookup is embedded in model state, so prediction needs no CoolProp or network.

$$q_{\text{ex}}(P, T) = \sum_{j=1}^2 A_j \frac{b_j(T)f(P, T)}{1 + b_j(T)f(P, T)} - 100V\rho_g(P, T),$$

$$b_j(T) = \exp[\ell_j + Q_j(1/T - 1/(160\text{K}))].$$

With f in bar, ρ in g/cm^3 , V in cm^3/g , the displacement term is wt.%. Coefficients are $A_1=1.15792$, $\ell_1=-3.407412$, $Q_1=627.7136$ K, $A_2=5.97028$, $\ell_2=-5.465494$, $Q_2=466.2582$ K, $V=0.870508$ cm^3/g . Q is an effective fitted thermal scale, not a measured adsorption enthalpy. V is estimated from uptake and is not an independent pore-volume measurement. Normal hydrogen is an explicit ortho/para assumption.

Whole-composition development MAE is 0.128481, versus a correctly specified ideal-density excess-Langmuir 0.141589 and real-gas one-site 0.152111. EG5 exposed diagnostic MAE is 0.201460, versus 0.208098 for the corrected ideal baseline and old 0.546458. Thus most headline improvement repairs the previous weak specification; the two-site gain over the strong ideal comparison is only 3.19% on the diagnostic. Global EG5 bias is -0.194047 wt.%, retained as a warning.

Curve-shape and operating-quantity falsifiers

At 77 K adsorption, the corrected model's continuous peak is within 3.91 bar of the sampled EG5 maximum, much closer than the old rule. At 160/273 K the predicted and sampled maxima are at the measured upper boundary: these are not identified interior optima. The 5-to-100 bar excess-uptake swing errors are 0.44658, 0.20761 and 0.03424 wt.% at 77/160/273 K. This is an excess-curve diagnostic, not independently measured absolute deliverable hydrogen. Desorption coverage differs and cannot supply matched-pressure causal hysteresis.

Pure graphite dilution, branch offsets and temperature-dependent capacity do not improve development sufficiently. Two-site effective parameters have similar fold ranges but appreciable conditioning; fitted sites do not establish physical pore populations. The publisher README describes PSD as MIL101/GO despite MOF5/EG filenames, preventing silent admission of its cumulative volume as independent hydrogen displacement calibration.

Value, limits and next evidence

The solid contribution is physically correct excess modeling, honest strong-baseline comparison and a much better bounded curve-shape prediction. This is known adsorption-family reproduction with a scoped retrospective approximation; no new pore mechanism, optimum-pressure law or deployed storage benefit is admitted. Independently documented pore/displacement volume, repeated freshly synthesized samples and matched operating cycles are needed for stronger claims.

Reproduction and evaluation

Run `python run.py` to verify frozen predictions. Use `python evaluator/evaluate.py example --output /tmp/submission` and `python evaluator/evaluate.py score --submission /tmp/submission --output /tmp/report --trust-code` in a new output location. This trusted-code replay is not a security sandbox. Evaluator scores exposed targets and does not grant novelty from error. Exact source/sample anchors are in `data/observations.csv.gz`; complete numerical evidence and unfavorable models remain in `evidence/`.

Sources and prior art

- Native MOF hybrid record
- Leachman hydrogen EOS
- Dual-site MOF hydrogen prior work

Targeted primary-source checking is not exhaustive novelty adjudication. Established model-family success is distinguished from a new physical law.